Formation 3 :

Evaluation du modèle

KRYPTOSPHERE®

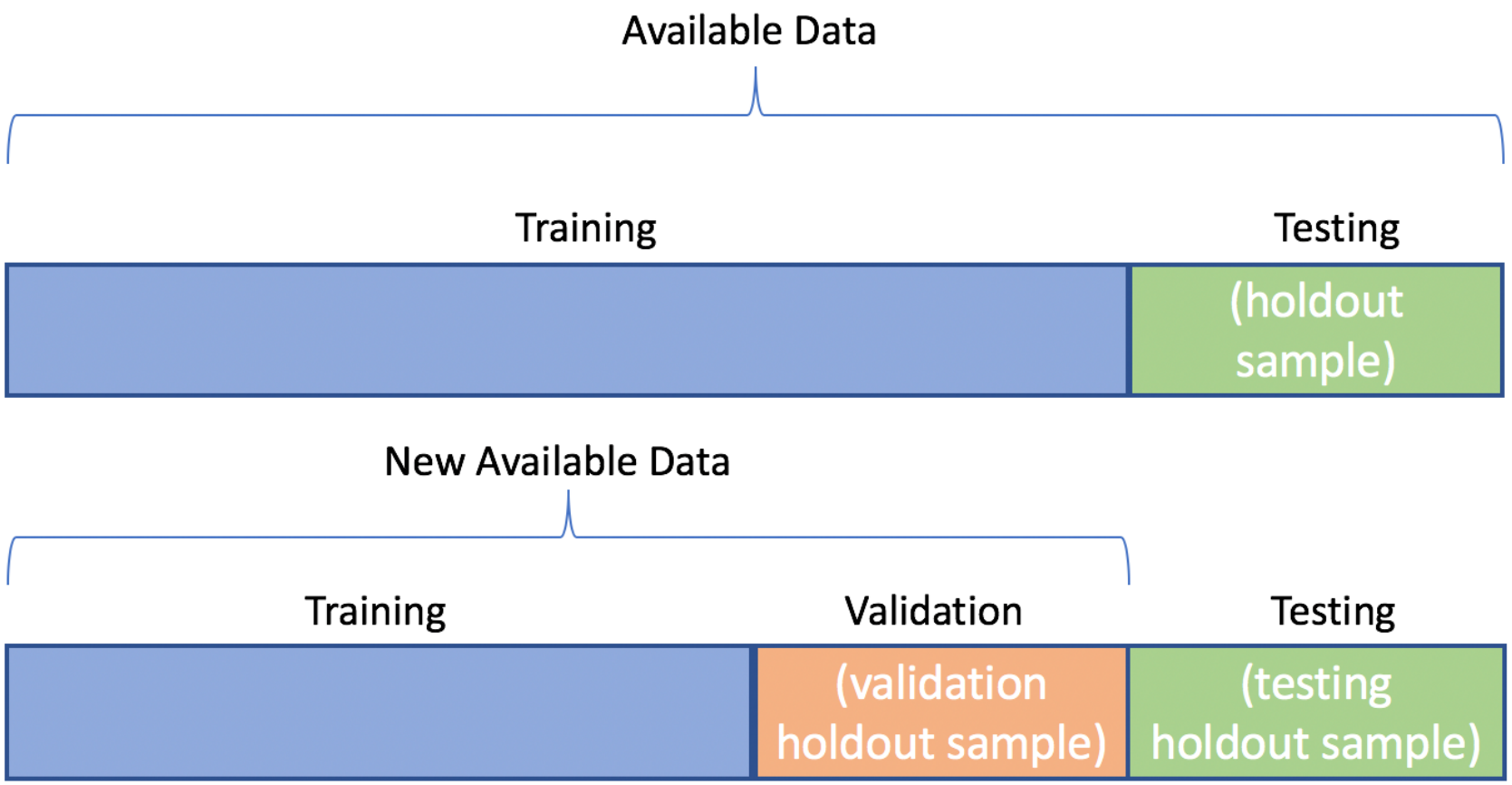
Plan :

Test set et Overfitting

Comment évaluer un modèle ?

Comment améliorer un modèle : les hyperparamètres

- Training set vs Test set

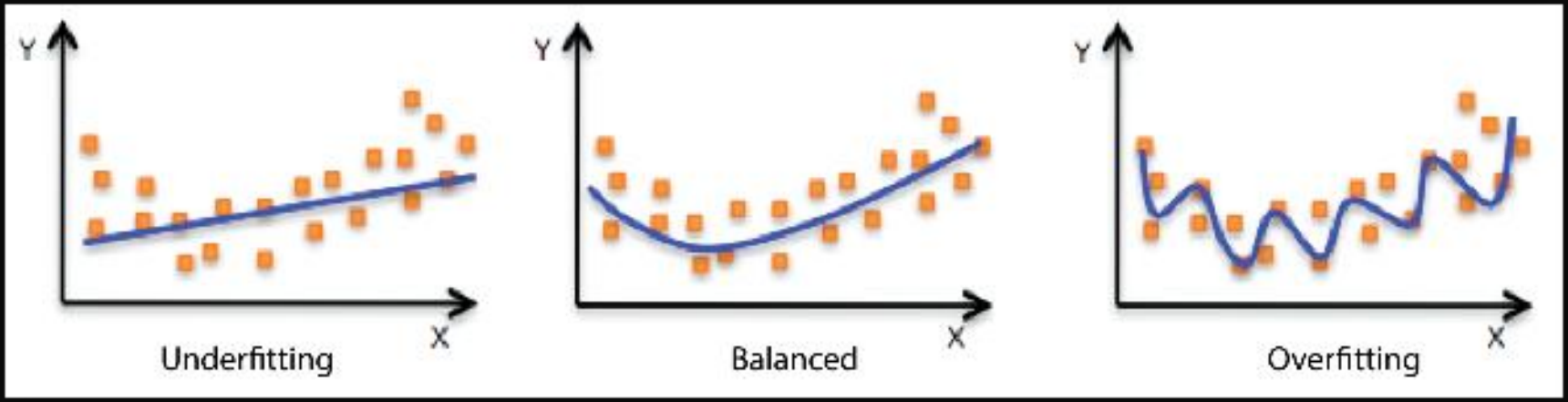


Séparer la dataset en python

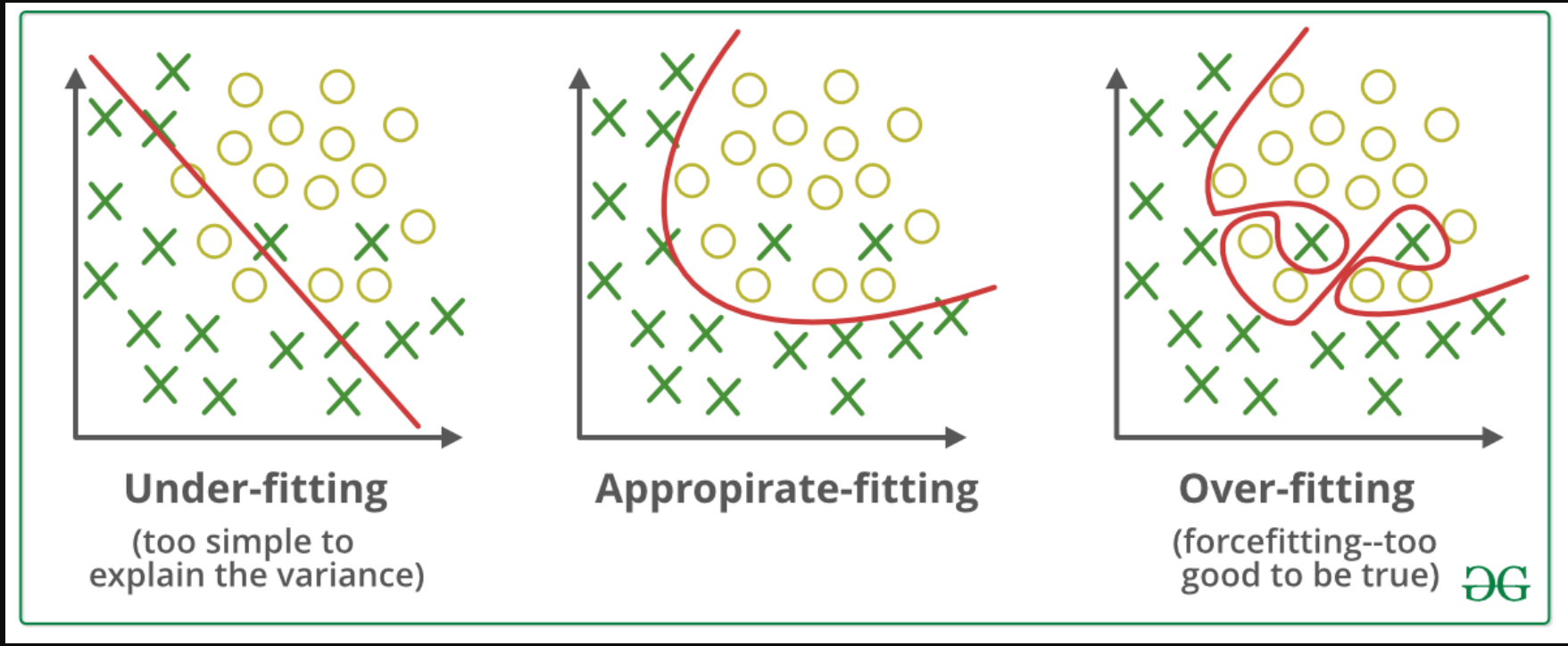
- Import sklearn

- from sklearn.model_selection import train_test_split

- X_train, X_test, y_train, y_test = train_test_split(
  X, y, test_size=0.2, random_state=42
)



Underfitting et Overfitting



Evaluer un modèle de régression : mean squared error

$$\mathrm{MSE} = \frac{1}{n} \sum_{i=1}^{n} (y_i - \tilde{y}_i)^2$$

Et en python?

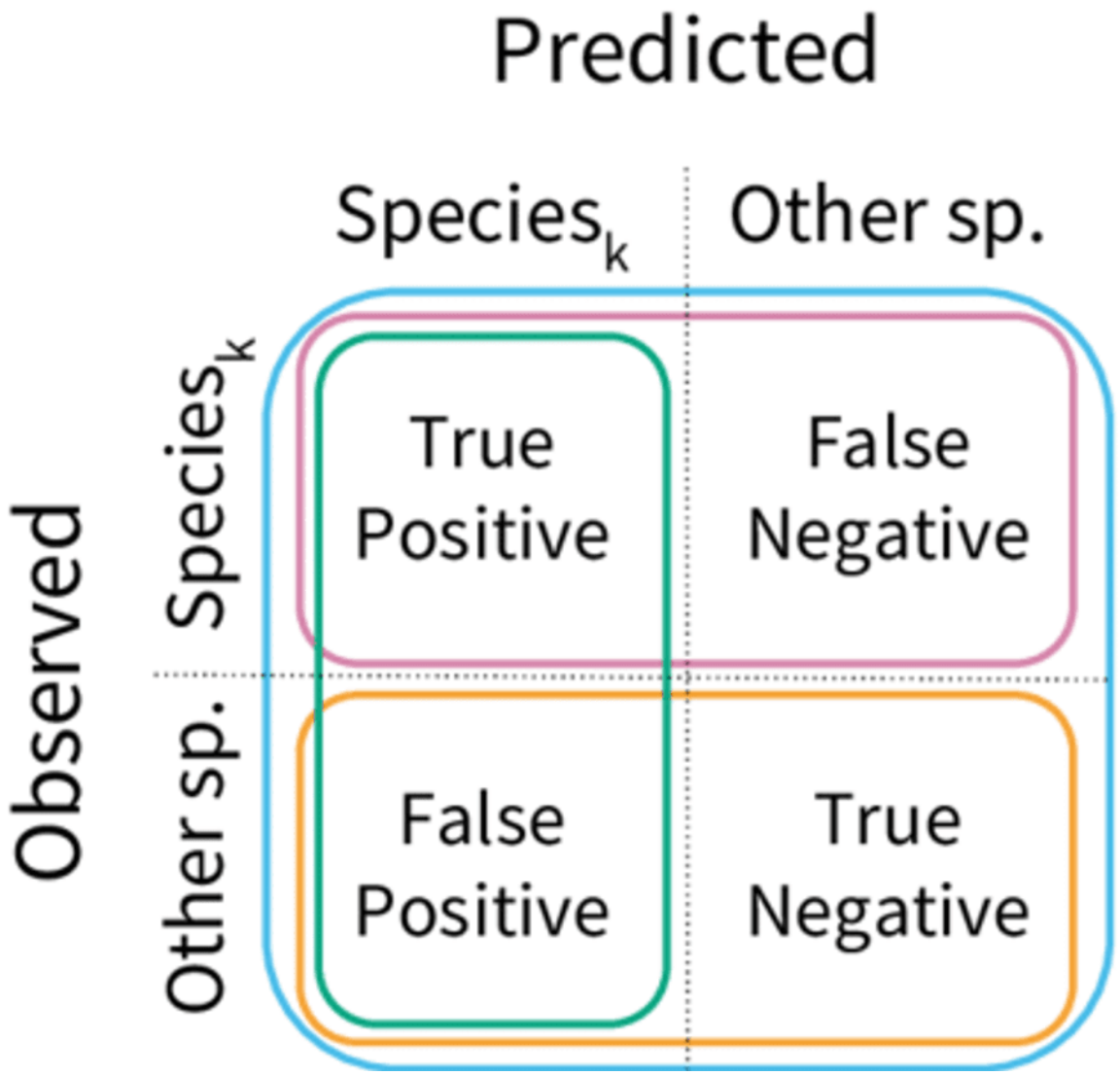
- from sklearn.metrics import mean_squared_error
- rmse = np.sqrt(mean_squared_error(y_test, y_pred))

Evaluer un modèle de classification : accuracy

$$\text{Accuracy} = \frac{\text{Correct predictions}}{\text{All predictions}}$$

Et en python?

- from sklearn.metrics import accuracy_score
- Accuracy = accuracy_score(y_test, y_pred)



$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

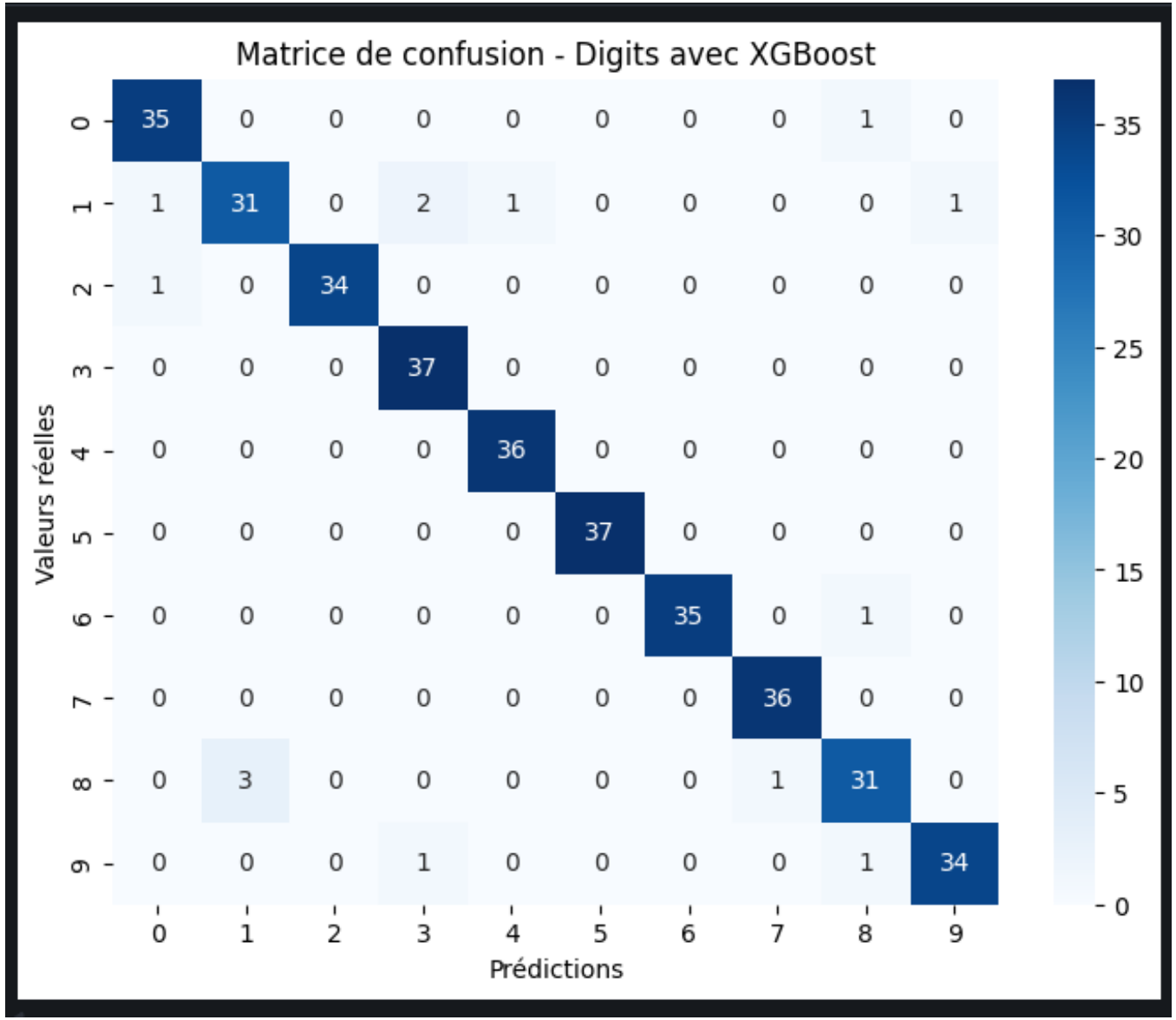
$$\text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall}$$

Et en python?

- from sklearn.metrics import classification_import
- classification_report(y_test, y_pred)

Confusion Matrix



Et en python?

```
cm = confusion_matrix(y_test, y_pred)

plt.figure(figsize=(8,6))
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues",
        xticklabels= ... ,
        yticklabels=... )
plt.xlabel("Prédictions")
plt.ylabel("Valeurs réelles")
plt.title("Matrice de confusion - Digits avec XGBoost")
plt.show()
```

Comment améliorer ces performances : les hyperparamètres

- Dans model = xgb.XGBRegressor(

  n_estimators -> nombre d’arbre (Attention overfitting)

  learning_rate

  max_depth -> profondeur max des arbres

)

Eviter l’overfitting : l’early stopping

- Besoins de 3 jeux de données : train, eval et test

- model.fit(X_train, y_train,

```
eval_set=[(X_valid, y_valid)],
eval_metric="rmse",
early_stopping_rounds=20,
verbose=True)
```

A vous de jouer !

Objectif : reprendre le TP Boston Housing, et évaluer votre modèle puis l’améliorer

- La correction de la semaine dernière est sur le drive